

// PROJECT PROPOSAL // GROUP 45



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// GROUP 45 - CYBER SHIELD — CAPSTONE PROJECT — 2026

CYBER SHIELD

HUMAN FIREWALL CHALLENGE

AI Scam Simulator + Immunity Trainer

PROJECT TYPE

Capstone Project

TRACK / SECTOR

Security and Safety

SDG ALIGNMENT

SDG 4 // 9 // 16

GROUP NAME

Group 45 - Cyber Shield

PROJECT ONE-LINER

"Cyber Shield doesn't just detect scams; it trains users to resist them."

CYBER SHIELD

// 01 PROBLEM STATEMENT

The Problem We Are Solving

Online scams are becoming increasingly sophisticated, targeting individuals through phishing emails, fake websites, social engineering messages, and fraudulent links.

While many cybersecurity tools focus on detecting scams, the human factor remains the weakest link. Employees, students, children, and elderly individuals often fall victim due to lack of practical awareness and real-world exposure.

Organizations also face major financial and reputational losses due to human error in handling cyber threats.

There is a need for an interactive and practical training solution that helps users recognize and resist scams before real attackers exploit them.

// 03 PROPOSED SOLUTION

What Is Cyber Shield?

Cyber Shield is an AI-powered interactive scam simulation platform designed to train users to identify and resist cyber threats.

Rather than simply detecting scams, the platform builds user immunity through realistic, hands-on experience.

KEY FEATURES

- AI-generated phishing emails and scam messages
- Fake login pages to test credential safety awareness
- Interactive escape-room style challenges
- Real-time feedback on user decisions
- Personalized cybersecurity improvement tips

Users learn by doing, making cybersecurity training engaging, practical, and memorable.

Roles and Responsibilities

A. TECHNICAL PROJECT MANAGER

(Best fit for Technical PM students)

RESPONSIBILITIES

- Plan project scope and objectives
- Create timeline and sprint plans
- Assign responsibilities and manage workflow
- Coordinate communication between technical teams
- Track risks and ensure timely delivery
- Prepare project documentation and reports

DELIVERABLES

- // Project roadmap
- // Task schedule
- // Risk management plan
- // Progress tracking dashboard

B. CYBERSECURITY ANALYST

(Cybersecurity students)

RESPONSIBILITIES

- Research modern cyber threats and scam patterns
- Design realistic phishing and scam scenarios
- Develop cybersecurity awareness content
- Identify user vulnerabilities
- Ensure platform follows security best practices

DELIVERABLES

- // Scam scenario database
- // Threat analysis report
- // User safety guidelines

C. SECURITY OPERATIONS AND RISK LEAD

(Cybersecurity students)

RESPONSIBILITIES

- Design the Human Risk Score framework
- Analyze how users respond to threats
- Map behavioral weaknesses
- Develop mitigation strategies
- Ensure ethical handling of user data

DELIVERABLES

- // Risk scoring model
- // Behavior analysis framework
- // Security compliance checklist

// 04 TEAM TASK SHARING (CONTINUED)

D. AI SYSTEMS COORDINATOR

(Cybersecurity student with coding OR Technical PM with tech skills)

RESPONSIBILITIES

- Plan AI feature implementation
- Coordinate between AI development and platform team
- Oversee scam scenario automation
- Manage AI workflow and testing
- Ensure AI outputs align with project goals

DELIVERABLES

- // AI feature plan
- // AI workflow diagrams
- // System integration documentation

E. TECHNICAL IMPLEMENTATION LEAD

(Cybersecurity student with coding background)

RESPONSIBILITIES

- Develop interactive simulation platform
- Build phishing email and fake login interfaces
- Integrate AI and scoring systems
- Manage backend functionality
- Ensure system reliability

DELIVERABLES

- // Functional application
- // Simulation engine
- // System integration

F. DOCUMENTATION AND PRESENTATION LEAD

(Technical PM students)

RESPONSIBILITIES

- Prepare project proposal and documentation
- Develop demo presentation slides
- Translate technical features into simple explanations
- Coordinate demo storytelling
- Prepare user guides

DELIVERABLES

- // Project documentation
- // Presentation slides
- // Demo script

SHARED RESPONSIBILITIES

Brainstorming features // Testing simulations // Demo rehearsal // Feedback improvements

// 05 AI COMPONENT

Artificial Intelligence Functions

SCENARIO GENERATION

AI creates realistic phishing emails, scam messages, and fraudulent web pages.

BEHAVIORAL ANALYSIS

AI monitors how users interact with simulations; click patterns, hesitation time, information entered.

RISK SCORING SYSTEM

AI calculates a Human Risk Score indicating how vulnerable a user is to scams.

PERSONALIZED TRAINING

AI recommends tailored safety tips and learning modules based on user weaknesses.

// 06 TARGET AUDIENCE

Who Is This Built For?

COMPANIES

Staff cybersecurity awareness and phishing resistance training.

SCHOOLS

Digital safety education integrated into curricula.

FAMILIES

Child online protection and parental scam guidance.

ELDERLY

Accessible scam awareness for the most vulnerable.

// **SEE PAGE 7** Full go-to-market strategy, budget, team costing & milestones in the Project Management Scope.

// 07 AI TOOLS & RESEARCH METHODOLOGY

How We Built This

This project was built using a combination of AI tools, combined research, and collaborative development, each contributing a unique strength to the final product.



PRIMARY DEVELOPMENT AI

Frontend development, game logic, Behavioural Analysis Engine, UI/UX design, documentation, and presentation structure. Primary AI used throughout the entire project build.



CONTENT & IDEATION

Brainstorming scam scenarios, refining problem statements, drafting initial proposal content and game narrative ideas. Used for early ideation and content generation.



RESEARCH & VALIDATION

Cross-referencing cybersecurity statistics, validating scam patterns, and supporting SDG alignment research across Africa. Used to verify facts and regional data.



DATA & STATISTICS

Sourcing cybercrime statistics, real-world scam examples, and best practices in cybersecurity awareness training. Foundation for all data referenced in this project.

■ COMBINED AI + HUMAN INTELLIGENCE APPROACH

Multiple AI tools were used collaboratively alongside team research and human insight, ensuring accuracy, realism, and impact.

// RESEARCH METHODOLOGY

- **Claude:** Used for all code generation, UI/UX design, documentation, PDF creation, and spreadsheet building. The primary engine behind the entire technical build.
- **ChatGPT:** Used in early ideation phases to brainstorm scam scenarios, draft content, and refine the problem statement narrative.
- **Google Gemini:** Used to cross-reference cybersecurity statistics specific to Africa, validate scam patterns, and confirm SDG alignment.
- **Google Search:** Used as the primary research engine to source real-world cybercrime statistics, phishing examples, regional scam data, and best practices in cybersecurity awareness training.

// 06 PROJECT MANAGEMENT SCOPE

Go-To-Market Strategy & Project Management

The following page presents the full Project Management Scope for Cyber Shield, developed as a companion document to this proposal. It outlines the complete go-to-market strategy, team structure, budget breakdown, 18-month delivery timeline, marketing campaign, revenue model, risk register, and key success milestones required to bring this platform to market.

BUDGET & COSTING

\$895,720 total investment across 18 months covering 14 roles, tech, marketing & contingency

TEAM & TIMELINE

14 specialists across 5 delivery phases over 18 months; Cybersecurity, AI, Dev, Legal & marketing

REVENUE & IMPACT

B2C, B2B SaaS, Government & White-label API. Year 2 Annual Recurring Revenue target: \$1.2M.

// SEE OVERLEAF - PAGE 7

The full Project Management Scope infographic is presented on the next page.

// THIS SECTION COVERS:

- 01 Team Professionals & Costing**
14 roles with monthly rates, durations and total costs
- 02 Project Timeline**
5-phase 18-month delivery roadmap to market launch
- 03 Full Budget Breakdown**
Personnel, infrastructure, marketing and contingency
- 04 Marketing Campaign Strategy**
Pre-launch, launch and growth phases with channels
- 05 Revenue & GTM Model**
B2C, B2B SaaS, government and white-label streams
- 06 Risk Register**
5 key risks with impact ratings and mitigations
- 07 Key Milestones**
7 success checkpoints from Month 3 to break-even

// 06 PROJECT MANAGEMENT SCOPE

Go-To-Market Strategy & Project Management

\$127,500

TOTAL BUDGET

18 MOS

GTM TIMELINE

14

PROFESSIONALS

\$35K

MARKETING

5

PHASES

// 01 TEAM PROFESSIONALS & COSTING

#	ROLE	DISCIPLINE	RATE/MO	DUR	TOTAL
01	Technical Project Manager Scope, sprints, risk, delivery	Project Mgmt	\$4,500	18 mo	\$81,000
02	Lead Cybersecurity Analyst Threat design, scam scenarios	Cybersecurity	\$5,200	12 mo	\$62,400
03	Security Operations / Risk Lead Human Risk Score, compliance	Cybersecurity	\$4,800	12 mo	\$57,600
04	AI / ML Engineer Behavioural engine, Claude API	AI Engineering	\$6,500	14 mo	\$91,000
05	Full-Stack Developer (Lead) React frontend, simulation engine	Software Dev	\$5,800	14 mo	\$81,200
06	React Native Developer iOS & Android app (Phase 5)	Mobile Dev	\$5,000	6 mo	\$30,000
07	UI/UX Designer Game UI, escape room, branding	Design	\$3,800	10 mo	\$38,000
08	Legal / Compliance Officer POPIA/GDPR, privacy policy	Legal	\$4,200	4 mo	\$16,800
09	Scrum Master / Agile Coach Sprint facilitation, velocity	Project Mgmt	\$3,500	18 mo	\$63,000
10	Cybersecurity Content Strategist Safety tips, scenario writing	Cybersecurity	\$2,800	10 mo	\$28,000
11	QA / Test Engineer Game testing, UAT	QA Engineering	\$3,200	8 mo	\$25,600
12	Digital Marketing Manager Campaigns, SEO, partnerships	Marketing	\$3,500	12 mo	\$42,000
13	DevOps / Cloud Engineer AWS/GCP, CI/CD, infrastructure	DevOps	\$4,500	12 mo	\$54,000
14	Business Development Lead B2B sales, school & corp partnerships	Business Dev	\$4,000	12 mo	\$48,000
TOTAL PERSONNEL (18 MOS)			TOTAL PERSONNEL (18 MOS) \$718,600		

// 02 PROJECT TIMELINE – 18 MONTHS TO MARKET

● PHASE 1

MONTHS 01-03

DISCOVERY & SCOPING

● PHASE 2

MONTHS 04-07

CORE BUILD

● PHASE 3

MONTHS 08-11

TESTING & BETA

● PHASE 4

MONTHS 12-15

MARKETING LAUNCH

● PHASE 5

MONTHS 16-18

MOBILE & SCALE

// 03 FULL BUDGET BREAKDOWN

PERSONNEL

Team Salaries (18 mos) \$718,600

Freelancers / Contractors \$18,000

TECHNOLOGY & INFRASTRUCTURE

Cloud / Hosting (AWS/GCP) \$14,400

AI API (Claude) Costs \$8,400

Software Licences / Tools \$6,000

MARKETING & GROWTH

Digital Campaigns (Meta/Google) \$12,000

Influencer / PR / Events \$8,000

Content Creation / Video \$7,500

SEO / Community / Socials \$7,500

OPERATIONAL & CONTINGENCY

Legal / Compliance / IP \$8,500

Admin / Office / Comms \$5,400

Contingency Reserve (10%) \$81,420

TOTAL PROJECT INVESTMENT (18 MOS)

\$895,720

BREAK-EVEN TARGET

MONTH 22

// 06 PROJECT MANAGEMENT SCOPE (CONTINUED)

Marketing · Revenue · Risk · Milestones

// 04 MARKETING CAMPAIGN STRATEGY

PRE-LAUNCH

MONTHS 10-11

Beta waitlist campaign, teaser content on LinkedIn/X/TikTok, cybersecurity influencer seeding, press kit to African tech media (TechCabal, IT Web SA). Target: 5,000 sign-ups before launch.

LAUNCH

MONTH 12

Product Hunt launch, live demo event (virtual + Cape Town/Lagos), paid Meta & Google campaigns, school partnership press release, challenge hashtag #HumanFirewall.

GROWTH

MONTHS 13-18

Corporate B2B outreach via LinkedIn Sales Navigator, school curriculum pilot grants, referral programme (earn badges for sign-ups), quarterly threat reports for PR, App Store / Play Store featured listing push.

META ADS

GOOGLE ADS

SEO/CONTENT

INFLUENCERS

PR/MEDIA

B2B OUTREACH

PRODUCT HUNT

COMMUNITY

// 05 REVENUE & GO-TO-MARKET MODEL

B2C FREE + PREMIUM

Free: 2 game levels. Premium: \$9.99/mo — all 5 levels, AI coaching, certificate, leaderboard. Schools: \$2/student/year (volume licensing).

B2B CORPORATE LICENSING

SaaS platform: \$499–\$2,999/mo per company tier (SME to Enterprise). Includes staff dashboards, HR risk reports, branded modules.

GOVERNMENT / NGO CONTRACTS

Grant-funded deployments via digital safety ministries, cybercrime bureaux, and UNICEF / AU digital inclusion programmes.

WHITE-LABEL / API

License Cyber Shield engine to banks, telcos and insurers for customer cyber-literacy compliance. \$15K–\$50K setup + rev share.

YR 2 REVENUE TARGET

\$1.2M ARR

// 06 RISK REGISTER

AI API cost overrun

Rate limiting + caching layer

HIGH

Data privacy breach

No real PII stored; encrypted sessions

CRITICAL

Talent retention

Equity options + flexible contracts

MED

Competitor entry

Africa-first moat + community lock-in

MED

Scope creep

Strict MoSCoW in each sprint

HIGH

// 07 KEY MILESTONES & SUCCESS METRICS

MO 03 Tech stack + legal entity registered

MO 07 MVP live with 5 levels + AI engine

MO 11 500 beta users — NPS target >= 65

MO 12 Public launch — 10K users month 1

MO 15 50 corporate clients signed

MO 18 Mobile app live — 100K+ total users

MO 22 Break-even — ARR approaching \$1.2M